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## Vehicle storage system


### Abstract

There is proposed vehicle storage apparatus, including a lifting jack having spaced apart lifting tines, a dolly being reversibly couplable to the lifting jack in a number of alternate positions, and a centre point device. The centre point device is used for determining the weight distribution along a longitudinal axis of the dolly when a vehicle is located thereon to assist in the suitable attachment of the dolly to lifting tines, to thereby inhibit tipping of the lifting jack when in the raised position or otherwise dislodgement of the vehicle stored thereon.



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| **Inventors:** | **Horvat; Danny Andrew (Botany, AU)** |
| **Applicant:** | **HORVAT NOMINEES PTY LTD** (Botany, AU) |
| **Family ID:** | **81600685** |
| **Assignee:** | **Horvat Nominees Pty Ltd (New South Wales, AU)** |
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*Primary Examiner:* Tran; Diem M

*Attorney, Agent or Firm:* Sand, Sebolt & Wernow Co., LPA

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## Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to an apparatus and method for vehicle storage and in particular for the storage of motorbikes.

BACKGROUND OF THE INVENTION

(2) Motorbikes are popular as a means of transport and are often collected by motoring enthusiasts. For many enthusiasts their motorbike remains in storage for the majority of the time. Some enthusiast may also have multiple motorbikes that may be used infrequently.

(3) This often entails the storage of the motorbike or multiple bikes in a garage or storage locker, which often have limited space. This may mean that the bikes need to be moved frequently to access other items being stored, which increases the risk of damaging the bike. In the case of

collectable motorbikes any damage can significantly affect the enthusiast's investment.
(4) There are various motorcycle bike lift benches and portable jack scissor lifts disclosed in the prior art, which can be used to lift a bike off a ground surface. This is typically done to assist a user in undertaking maintenance or repair. One such device is disclosed in International Application No. PCT/US2004/023641 (Thurm) that teaches a portable motorcycle lift that utilizes a scissor frame configuration. However, the bike is only lifted off the ground a small distance and the scissor frame obstructs access under the device for storage purposes.
(5) There are storage racks available wherein motorbikes or other types of small land or water vehicle, can be positioned on a pallet and placed onto shelving. This however requires the use of a forklift, which in many cases may not be available in typical storage situations. Furthermore, these systems are large and typically are not suitable for garages or storage lockers. Accordingly, such systems are typically not suitable for most motorbike owners.
(6) There are also various dolly devices disclosed in the prior art, which can be used to elevate, at least one wheel of a motorbike off the ground, such that the bike can be more easily manoeuvred into tight spaces.
(7) One such dolly is disclosed in U.S. Patent Application No. 20020109130, in the name of TC Development and Design, Inc., which teaches a portable motorcycle lift that permits the elevation of the front and rear ends of the motorcycle independently of one another. Another dolly system is disclosed in International Application No. PCT/AU2015/000428, in the name of Black MC Pty Ltd, which discloses a dolly frame being supported on multidirectional wheels. The dolly frame can be pivoted to engage a tyre of a motorbike and then lowered to raise a wheel of the bike to assist in manoeuvring the vehicle for storage.
(8) These systems do not however enable the utilisation of the vertical space within a storage area to assist in storage of one or more motorbikes.
(9) The term “vehicle” used throughout the description and claims should be given its broadest definition and includes motorbikes, quad bikes, motorised trikes or other types of small motorised land vehicles or watercraft, such as but not limited to, jet skis and other personal watercraft (PWC), which are also often stored in garages or storage lockers.
(10) It should be appreciated that any discussion of the prior art throughout the specification is included solely for the purpose of providing a context for the present invention and should in no way be considered as an admission that such prior art was widely known or formed part of the common general knowledge in the field as it existed before the priority date of the application.
SUMMARY OF THE INVENTION
(11) It is an object of the present invention to provide an apparatus for use in the storage of a vehicle, at a height above the ground, which is configured to inhibit tipping thereof. Further objects of the present invention are to overcome at least some of the aforementioned problems, or at least provide the public with a useful alternative. The foregoing objects should not necessarily be considered as cumulative and various aspects of the invention may fulfil one or more of the above objects.
(12) In one aspect of the invention, but not necessarily the broadest or only aspect, there is proposed a vehicle storage apparatus, including:
(13) a lifting jack having spaced apart lifting tines;
(14) a dolly being reversibly couplable to the lifting tines in a number of alternate positions; and
(15) a centre point device for determining the weight distribution along a longitudinal axis of the dolly when a vehicle is located thereon, to assist in the selection of one of the number of alternate positions for the suitable attachment of the dolly to the lifting tines, to thereby inhibit tipping of the lifting jack when the lifting tines are in the raised position or otherwise dislodgement of the vehicle stored thereon
(16) The reader will appreciate that the centre point or fulcrum of the dolly will vary depending upon the vehicle being stored thereon and the particular configuration of the vehicle, which may

vary over time.
(17) Preferably, the dolly is movable laterally relative the lifting tines of the lifting jack and reversibly couplable to the lifting lines, once a longitudinal centre point of the dolly with the vehicle positioned thereon is determined.
(18) The dolly may be coupled to, or couplable to the lifting tines, using a nut and bolt that are engageable through co-axially alignable holes, or a clamp mechanism, or a clip may be provided such that the dolly can be coupled to, or couplable to the lifting jack in a variety of positions depending upon the weight distribution of the vehicle being supported on the dolly.
(19) The dolly and centre point device are preferably separate devices. However, in an alternate form, the centre point device may form part of the dolly and may be moveable between a downwardly extending position when it is being used to determine the weight distribution and a disengaged or retracted position when the vehicle is being supported on caster wheels or sliders of the dolly. In still another form the caster wheels or sliders may be retractable to expose the centre point device such that it can be used to determine weight distribution, along the longitudinal axis or length of the dolly.
(20) As the reader will appreciate the longitudinal centre point will change depending upon what vehicle is being stored. Furthermore, the centre point of a particular vehicle may change depending upon the amount of fuel in the fuel tank, or if accessories are added. Accordingly, the dolly is reversible fixable to the lifting tines so that the longitudinal centre point can be reassessed and the dolly can be repositioned prior to storage.
(21) In another aspect of the invention there is proposed a motorbike storage apparatus, including:
(22) a dolly having a plurality of generally vertical primary holes extending therethrough, the dolly being supportable on wheels or slides, and configured to hold a motorbike thereon;
(23) a lifting jack including spaced apart lifting tines, each of the lifting tines including at least one generally vertical secondary hole extending therethrough; and
(24) a centre point device, being configured to selectively support the dolly thereon, for use in determining a weight distribution along a longitudinal axis of the dolly, whereby at least two of the vertical primary holes are identified for coaxial alignment with respective secondary holes extending through the lifting tines, to thereby enable passage of a respective fixing means therethrough, wherein the dolly is fixedly connectable to lifting tines to thereby provide a generally even distribution of weight to inhibiting tipping of the apparatus or otherwise dislodgement of the vehicle stored thereon.
(25) The lifting jack may be a mobile unit having lockable caster wheels, ground engaging members with rollers, a vertical frame and lifting mechanism for raising the spaced apart lifting tines.
(26) Alternatively, the lifting jack may comprise a frame attachable to a wall or other vertical support, and lifting tines movably connected thereto.
(27) The lifting mechanism may include a hydraulic or pneumatic system for assisting in the raising of the dolly/motorbike. The lifting mechanism may also include chains, tracks and other components that would be obvious to a person skilled in the art, and which will not be described in detail.
(28) The apparatus preferably includes a locking member, which is reversibly engageable to hold the lifting tines in a raised positioned. The locking member may inhibit unintentional lowering of the lifting tines when the motorbike is being supported thereon. As the reader will appreciate, this is important to prevent the lifting tines coming into contact with an object being stored or positioned below the lifting tines, such as another motorbike or vehicle.
(29) Preferably, the lifting jack includes a wheel lock or locks for the caster wheels. In one form, the wheel lock or locks may be automatically activated when the lifting tines are raised. This thereby prevents the lifting jack from being moved when the lifting tines are in the raised position. If the lifting jack were to be moved when a motorbike was being stored at a height there would be

an increased risk that the apparatus may topple over due to the weight distribution or sideways pressure being applied.
(30) In one form, the dolly comprises a plank being affixed to two spaced apart cross members. Each cross-member includes a caster wheel assembly at opposite ends thereof, such that the plank is effectively supported on four caster wheels. Each cross-member may further include tie-down eyelets for the attachment of ratchet straps or the like, to assist in securing the motorbike when being stored.
(31) The plank may include a movable ramp at a first end, and a wheel engaging member at an opposite second end. Preferably, the plank is generally L-shaped and comprises an upwardly open channel adjoining a vertical end stop at the second end.
(32) In one form the wheel engaging member may comprise a rearwardly open support member, which is fixedly attached to the vertical end stop of the plank. The rearwardly open support member is configured to retain a front portion of the front wheel of the motorbike. The wheel engaging member may further include a pivotable support member which is held at least partly within the upwardly open channel. In use as the motorbike is rolled up the ramp and along the upwardly open channel of the plank, the front wheel rolls over the pivotable support member and abuts the rearwardly open support member. As the reader should appreciate the pivotable support member, pivots as the wheel rolls thereover, such that it engages a rear portion of the front wheel. The plank may also include a non-slip surface to assist in holding the front wheel in place, when positioned within the wheel engaging member.
(33) Furthermore, four threaded eyelet bolts may be positioned on either side adjacent the front wheel being configured to engage through holes in the plank. In one form the reader will appreciate that a three point wheel locking assembly is envisaged and provided to hold a front wheel.
(34) It is envisaged that the front wheel of the motorbike will engage the wheel engaging member.
(35) The pivotable support member may be repositioned along the upwardly open channel of the plank to accommodate different diameter wheels.
(36) The wheel engaging member may also comprise a part of the plank which is modified or attached thereto, to assist in retaining the wheel of the motorbike.
(37) The plurality of primary holes extended vertically through the plank of the dolly.
(38) A guide member may be slidably coupled to or configured to engage with the plank. In a preferred form, the guide member clips over an edge of the upwardly open channel and is configured to slide therealong. The guide member may be used in conjunction with the centre point device to determine which of the plurality of primary holes are to be used to coaxially align the at least one secondary hole through the lifting tines.
(39) The centre point device in one form comprises a roller having ends being enlarged to abut respective sides of the plank. In this way the centre point device can be placed under the plank and the dolly can be moved backwards and forwards in a longitudinal direction to determine the weight distribution and lateral centre point line of the dolly with motorbike supported thereon. The enlarged ends of the centre point device ensure corrected placement under the plank and inhibit the centre point device from rolling sidewardly out from under the dolly when in use.
(40) The diameter of the centre point device is preferably greater than the distance between the underside of the plank and the caster wheels such that the caster wheels are raised off the ground when the plank is supported on the centre point device.
(41) Once the dolly has been moved such that the centre point device is positioned such that there is even distribution of weight on either side, the guide member can be moved whereby a centre line of the guide aligns and is positioned vertically above an axis of rotation of the centre point device. In one form, the length of the guide member is the same as the distance between the secondary holes of the adjacent lifting tines. This means that the user can then mark or take note of the primary holes adjacent each end of the guide member, which can be used to attach the dolly to the lifting tines, to thereby evenly distribute the weight of the dolly/bike.

(42) An L-shape bracket may be used to assist in the attachment of the dolly to the lifting tines, in that it can be used as a back support for corrected positioning of the plank relative to the lifting tines so that the holes can be more easily aligned.
(43) In another aspect of the invention there is proposed a method of storing a motorbike, including the steps of:
(44) providing a motorbike storage apparatus, in accordance with any one of the above aspects, including, a dolly reversibly couplable to a lifting jack in a number of alternate positions, and a centre point device;
(45) positioning the motorbike on a plank of the dolly;
(46) lifting the dolly using the lifting jack;
(47) positioning the centre point device under the plank of the dolly;
(48) lowering the dolly using the lifting jack until the plank rests on the centre point device;
(49) moving the dolly in a longitudinal direction to determine generally the centre point of the motorbike and dolly, such that the weight is even distributed on either side of the centre point device;
(50) marking or otherwise noting the centre point/fulcrum and other related measurements on the plank of the dolly;
(51) lifting the dolly off the centre point device using the lifting jack;
(52) removing or moving the centre point device;
(53) using the markings of the centre point or other related measurement, to fix the plank of the dolly to the lifting jack, such that when the dolly and motorbike are raised by the lifting jack the motorbike storage apparatus is inhibited from tipping or otherwise dislodgement of the vehicle stored thereon.
(54) The above method preferably includes the step of securing the motorbike to the dolly, using at least one wheel engaging member, and/or clamps, and/or straps and/or other reversible attachable member for holding the motorbike in place relative to the dolly.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate an implementation of the invention and, together with the description and claims, serve to explain the advantages and principles of the invention. In the drawings,
(2) FIG. **1** is a front perspective view of one embodiment of a lifting jack of the motorbike storage apparatus of the present invention;
(3) FIG. **2** is a rear perspective view of the lifting jack of FIG. **1**;
(4) FIG. **3** is a perspective view of a dolly, centre point device, and guide member of the motorbike storage apparatus;
(5) FIG. **4** is an exploded view of the dolly of FIG. **3**;
(6) FIG. **5** is a top view of the dolly with guide member attached and centre point device of FIG. **3**;
(7) FIG. **6** is a front view of the lifting jack of FIG. **1** with motorbike positioned on the dolly;
(8) FIG. **7** is a side view of FIG. **6**, illustrating the motorbike positioned on the dolly and adjacent lift device;
(9) FIG. **8** is a front view of FIG. **7**, illustrating the motorbike and dolly positioned over the tines of the lift device;
(10) FIG. **9** is the front view of FIG. **8**, illustrating the tines of the lifting jack engaging and lifting the dolly and motorbike;
(11) FIG. **10** is the front view of FIG. **9**, illustrating the tines of the lifting jack in a lowered position and the dolly resting on the centre point device;

(12) FIG. **11** is the front view of FIG. **10**, illustrating the repositioning of the dolly such that the weight is evenly distributed and movement of the guide member;
(13) FIG. **12** is a top view of the dolly of FIG. **11**, with bike removed, illustrating the selection of the appropriate fixing holes;
(14) FIG. **13** is a perspective view of FIG. **12**, illustrating attachment of bolts;
(15) FIG. **14** is an enlarged top view of the area indicated on the FIG. **13**, showing attachment of the brackets and the position of the mounting holes of the brackets and the dolly affixed to the tines;
(16) FIG. **15** is a front view of the apparatus of FIG. **11**, illustrating the dolly attached to the tines of the lifting device;
(17) FIG. **16** is a side view of apparatus and motorbike of FIG. **15** illustrating the use of ratchet straps;
(18) FIG. **17** is an opposite side view of FIG. **16**;
(19) FIG. **18** is a side view of the apparatus of FIG. **17**, illustrating the lifting jack in a raised arrangement with a vehicle partly positioned under the raised dolly and motorbike;
(20) FIG. **19** is a front view of the apparatus illustrating two motorbikes stacked one above another, and supported on a respective dolly;
(21) FIG. **20** is a rear view of another embodiment of the lifting jack;
(22) FIG. **21** is a side view of the lifting jack of FIG. **20**;
(23) FIG. **22** is a top view of the lifting jack of FIG. **20**;
(24) FIG. **23** is a perspective view of another embodiment of the plank of the dolly;
(25) FIG. **24** is a reverse perspective view of the plank of FIG. **23**;
(26) FIG. **25** is a side view of the plank of FIG. **23**, illustrating the configuration prior to folding;
(27) FIG. **26** is a side view of the plank of FIG. **23**, after folding;
(28) FIG. **27** is an underside view of the plank of FIG. **23**, illustrating the position of the holes;
(29) FIG. **28** is an end view of the vertical end stop of the plank of FIG. **23** illustrating the position of the holes;
(30) FIG. **29** is a cross-sectional view through A-A of the plank of FIG. **24**, illustrating the configuration of the holes;
(31) FIG. **30** is a partial top view of the ground engaging members of the lifting jack of FIG. **20**, illustrating an enlargement of the roller configuration;
(32) FIG. **31** is a side view of the lifting tines of the lifting jack of FIG. **20**;
(33) FIG. **32** is an underside view of the lifting tines of FIG. **31**, illustrating the position of the plank fixing holes and corresponding nuts;
(34) FIG. **33** is a cross-sectional view through B-B of the lifting tines of FIG. **32**; and
(35) FIG. **34** is a top view of the lifting tines of FIG. **31**, indicating the attachment of two adjacent planks for supporting respective bikes thereon.

DETAILED DESCRIPTION OF THE ILLUSTRATED AND EXEMPLIFIED EMBODIMENTS

(36) Similar reference characters indicate corresponding parts throughout the drawings. Dimensions of certain parts shown in the drawings may have been modified and/or exaggerated for the purposes of clarity or illustration.
(37) Referring to the drawings for a more detailed description, there is illustrated a motorbike storage apparatus **10**, demonstrating by way of examples, arrangements in which the principles of the present invention may be employed.
(38) As illustrated in FIGS. **1** and **2**, the motorbike storage apparatus **10** includes a lifting jack **12** being in the present embodiment a mobile unit, having lockable caster wheels **14**, ground engaging members **16** with rollers **18**, a vertical frame **20**, a lifting mechanism **22**, handle actuator **23**, including a hydraulic cylinder valve, spaced apart lifting tines **24** having secondary holes **25***a* and **25***b*, handles **26** and a brake actuator **28**. The lifting jack **12** is of a configuration that is known in the prior art and will not be described in detail, since the skilled addressee will understand the

configuration and operation of the device.
(39) Turning to FIGS. **3** and **4**, there is illustrated a dolly **30** which is couplable or attachable to the lifting tines **24** of the lifting jack **12**. The dolly **30** of the present embodiment comprises a plank **32** being affixed to two spaced apart cross members **34**, **36**, by way of bolts **38** and nuts **40**. Each cross-member **34**, **36** includes a caster wheel assembly **42** at opposite ends thereof, such that the plank **32** is supported on the four lockable caster wheels **44**.
(40) Each cross-member **34**, **36** further includes tie-down eyelets **46** for the attachment of ratchet straps **48**, as illustrated in FIGS. **15** to **17**, to assist in securing the motorbike **50** to the apparatus **10** when being stored. The ratchet straps **48** may be attached as soon as the motorbike **50** has been positioned on the plank **32**, or at least prior to raising the motorbike **50** to a height above the ground.
(41) The cross-members **34**, **36** may be constructed from aluminium box section and have a length of 498 mm and a cross-sectional profile of 50 mm×25 mm. The cross-members **34**, **36** are mounted at either end of the plank **32** and bolted directly through selected 12 mm holes **92**.
(42) At each end of the cross-member **34**, **36** respective 95 kg rated multi directional trolley wheels/caster wheel assembly **42** are positioned with brakes fitted. The total height off the ground from the base of the plank **32** in the present embodiment is 95 mm, providing 5 mm clearance above the members **16**, **24** of the apparatus **10**. This is so the plank **32** is able to be manoeuvred interposition above locating bolt holes **92** on lifting tines **16** ready for lifting. The trolley wheels **42** are bolted onto the aluminium section **34** or **36** using 10 mm eyelet bolts **46** which provide securing points for safety ratchet straps **48** used to secure a bike **30** on to the plank **32**.
(43) The plank **32** has a ramp **52** attachable at a first end **54**, by way of a screw/bolt **56**. The first central hole **92** at entry of plank **32**, is in the present embodiment positioned 25 mm centre from the leading-edge and is done to accommodate the detachable ramp **52** for loading and unloading a bike or vehicle.
(44) The opposite second end **58** of the plank **32** includes a wheel engaging member **60**. The wheel engaging member **60** comprises a rearwardly open support member **62**, which in the present embodiment is attached at the second end **58** of the plank **32**, by way of pin **64** which extends through apertures **66**.
(45) The support member **62** is configured to retain a front of the wheel **68** of the motorbike **50**, as shown in FIG. **6** and is able to pivot slightly around pin **64** to compensate for different sized wheels **68**.
(46) The wheel engaging member **60** further includes a pivotable wheel capturing member **70**, which is held at least partly within the upwardly open channel **72** of the plank **32**. The pivotable wheel capturing member **70** is held in place by way of pin **74**, which is configured to alternatively engage cooperating apertures **76***a*, **76***b*, **76***c*, depending upon the size of the wheel **68**.
(47) In use, the four caster wheels **44** are locked and the motorbike **50** is rolled up the ramp **52** and along the upwardly open channel **72** of the plank **32**, in the direction of the arrow in FIG. **6**. The front wheel **68** rolls over the pivotable wheel capturing member **70** and abuts the rearwardly open support member **62**. As the reader should appreciate, the pivotable support member pivots as the wheel rolls thereover such that it engages a rear of the wheel, once it has passed thereover. The plank **32** may also include a non-slip surface **71** to assist in holding the front wheel in place, which positioned within the wheel engaging member or bolts that pass through selected holes **92** on either side of the front wheel to thereby hold the wheel in place.
(48) In the present embodiment the plank **32** is formed by a length of open channel which is cut at broken line **78**, and folded to form the upwardly open channel **72** and adjoining vertical end stop **80**, with an upper end **82** that is cut and folded. In the present embodiment L-shaped bracket **84** are attached using nuts **86** and bolts which engage apertures **90**, to thereby fix the vertical end stop **80** relative to the upwardly open channel **72**.
(49) The height of the vertical end stop **80** is in the present embodiment 420 mm, and is configured

to stop the bike moving forward and is also used to strengthen the plank **32** as it holds the edge of the front tyre in place at **62**. The upper end **82** vertical end stop **80** acts as a stiffener and may include indicia thereon to identify the source of the invention. The plank **32** is bent at a 90 degree angle and welded along the 45 degrees outline **78**.

(50) The upwardly open channel **72** and end stop **80** further includes a plurality of primary holes **92**, extended vertically through the plank **32**, the purpose of which will be described in detail below.

(51) As illustrated in FIG. **3**, the apparatus **10** further includes a guide member **94** that can be clipped over an edge of the upwardly open channel **72**, such that it is slidable therealong, as indicated by the arrows in FIG. **5**. The guide member **94** has a centre marking **96** and a length **98** that equals the distance between the secondary holes **25***a* or **25***b* through the adjacent lifting tines **24**. In one embodiment, the length is 525 mm, however the reader will appreciate that other length guide members could be used depending upon the distance between the secondary holes **25***a* or **25***b*.

(52) As indicated in FIG. **5**, the ends of the guide member **94** indicate which primary holes **92***a* and **92***b*, should be used to align with the secondary holes **25***a* or **25***b* in the respective lifting tines **24**.

(53) The apparatus **10** further includes a centre point device **100** comprising a cylindrical roller portion **102** and disc shaped ends **104** being larger than the cylindrical roller portion **102**. In this way when the plank **32** is positioned on the cylindrical roller portion **102** the shoulder **106** engages with respective vertical sides of the plank **32**. The shoulders **106** of the centre point device **100** ensure corrected placement under the plank **32** and inhibits the centre point device **100** from rolling sidewardly out from under the dolly **30**. The reader will appreciate that the opposite ends of the centre point device **100** may simply be enlarged to inhibit disengagement from under the plank **32**.

(54) The guide member **94** is used in conjunction with the centre point device **100** to determine which of the plurality of primary holes **92** are to be used to coaxially align the secondary holes **25***a* or **25***b* to be used to attach the plank **32** of the dolly **30** to the lifting tines **24**.

(55) In the present embodiment, as illustrated in FIG. **3**, the apparatus **10** further includes placement brackets **110** including a plurality of holes **112** to assist in aligning the primary and secondary holes **25***a* or **25***b*, **92**, as will be described with respect to FIG. **12**. In the present embodiment, the brackets **110** are held in place by bolts **114** that pass through holes **112** and apertures **25***a* or **25***b* in the respective lifting tines **24**.

(56) FIG. **7**, illustrates the step in using the apparatus **10**, wherein the dolly **30** is positioned in front of the lifting jack **12**. The dolly **30** is then moved in the direction of the arrow in FIG. **7** such that it is positioned over the top of the lifting tines **24**, as illustrated in FIG. **8**.

(57) The lifting tines **24** are then operated to raise the dolly **30** and motorbike **50** off the ground **118**, as indicate by the arrows in FIG. **9**, such that the caster wheels **44** are spaced apart from the ground **118**. The centre point device **100** can then be placed directly under the plank **32** in approximately the central region of the elongate length of the plank **32**.

(58) The centre point device **100** is then placed under the plank **32** and the lifting tines **24** are operation to lower the dolly **30**, as indicate by the arrows in FIG. **10**, such that the plank **32** is supported on the centre point device **100**. The dolly **30** with bike attached thereto can then be moved backwards and forward until the weight is evenly distributed on either side of the centre point device **100**, as illustrated in FIG. **11**.

(59) The guide member **94** can then be moved longitudinally in the direction of the arrows along the plank **32**, as indicated in FIG. **11**, such that the centre line **96** aligns and is positioned vertically above, the axis of rotation **120** of the centre point device **100**.

(60) The lifting tines **24** can then be used to again raise the dolly **30** to remove the centre point device **100** and then lowered again such that the motorbike **50** can be wheeled off the plank **32**, if required. As illustrated in FIG. **12**, the brackets **110** can then be attached to the lifting tines **24** to assist in aligning the primary holes **92***a*, **92***b*, which are select because they are adjacent ends of the

guide member **94**, with the underlying secondary holes **25***a* or **25***b* in the lifting tines **24**. The attachment of the brackets **110** means that the side of the plank **32** can be pushed up against the vertical portion of the bracket **110**, such that the plank **32** can be moved sidewards, backwards and forwards, to align the primary and secondary holes **92**, **25***a* or **25***b*. The reader should however appreciate that the use of the bracket **110** is not essential.
(61) FIG. **12** also illustrates the position of the axis of rotation **120** of the centre point device **100**, which aligns the centre line **96** of the guide member **94**. FIGS. **13** and **14** illustrate the use of bolts **122** to fix the plank **32** to the lifting tines **24**. It will be appreciated that if a different motorbike is to be stored or the weight distribution of the motorbike changes, due to variation in fuel within the tank or the addition/removal of accessories, the bolts **122** can be removed and the centre point reassessed using the above method.
(62) As illustrated in FIGS. **15** to **17**, ratchet straps **48** can be used to secure the motorbike **50** to tie-down eyelets **46**. Furthermore, ratchet straps **48** may also be used to secure the motorbike **50** to tie-down eyelets **124**. These ratchet straps **48** assist in keeping the motorbike **50** upright and engaged with the wheel engaging member **60**.
(63) The caster wheels **14** are then locked using the brake actuator **28**, to prevent it being moved when the vehicle is in the raised position. The lifting jack **12** can then be used to raise the motorbike **50** to a height above the ground, wherein a part of a vehicle **126** can be positioned under the raised motorbike, as illustrated in FIG. **18**. Alternatively, another motorbike **50***a* positioned on a respective dolly **30***a* can be moved under the raised motorbike **50**, as illustrate in FIG. **19**. A fluid catchment tray/s (not shown) can be placed between the bottom of the motorbike **50** and underlying motorbike **50***a* to prevent oil or other fluids dropping onto the motorbike **50***a*, or articles stored therebelow.
(64) The space under the raise motorbike **50** can also be used to store any other appropriately sized object.
(65) The reader will also appreciate that the centre point device **100** is used to determine the approximate longitudinal centre of gravity of the dolly **30** with the motorbike **50** positioned thereon. Accordingly, the dolly **30** can be affixed to the lifting tine **24** such that the weight is distributed evenly to inhibited the apparatus **10** from tipping when in a raised arrangement, as illustrated in FIGS. **18** and **19**.
(66) FIG. **18** also illustrates the jack safety chain **128** attached to the eyelet **124** using a shackle **130**, to thereby prevent the unauthorised lowering of the lifting tines **24**.
(67) FIGS. **20** to **22** illustrate a preferred embodiment of the lifting jack **12** with an elongate handle actuator **23** of the lifting mechanism **22**.
(68) The plank **32**, as illustrated in FIGS. **24** to **29**, may be formed by bending a length of steel having a plurality of primary holes **92**, extended therethrough. As illustrated in FIG. **25** the length of steel, which may be 2500 mm in length, is cut at **78** to enable bending. The sides of the upwardly open channel **72** are cut in a V-shape and then folded to form the L-shaped plank **32**, as illustrated in FIG. **26**.
(69) The sides of the upwardly open channel **72** at the entry point E or start of plank **32**, as illustrated in FIG. **23**, are cut at 45 degree on the side stiffener walls to inhibit damage to the bike tyres when loading the bike onto the plank **32**.
(70) The overall plank **32** length of the of the present embodiment of the folded device, as illustrated in FIG. **26** is in the present embodiment 2080 mm. This length of the plank **32** accommodates a range of different length bikes and when a bike is positioned on the plank **32** both front and rear wheels are not going over the elevation at points **52** and **70** at the same time. The usable length is reduced by the rearwardly open support member **62**, which in a preferred embodiment is a large rubber chock, to a length of 2018 mm. Although, the reader will appreciate that other length planks **32** are possible.
(71) As further illustrated in FIG. **26**, a rubber stop R may be attached to the leading edge of the

vertical end stop **80**. The rubber stop R is used as a rest for abutment against a wall or other vertical surface, and assists a user when loading a bike onto to the plank **32**, since it inhibits the unit from moving forwards.

(72) As illustrated in FIGS. **27** to **29**, the plurality of primary holes **92** extend through the plank **32**. The number of holes **92** and configuration of the preferred embodiment enable the plank **32** to use used on various sizes and configuration of bikes.

(73) The width of the plank **32** in one embodiment is 300 mm overall and 250 mm inside diameter between the inner edges of the upwardly open channel **72**, as illustrated in FIG. **29**.

(74) Four large 12 mm threaded eyelet bolts may be secured through selected holes **92** on either side of the wheel to hold the bottom edge of the front wheel of the bike as part of the wheel holding or locking mechanism.

(75) As further illustrated in FIGS. **27** and **28**, the hole pattern of the plank **32** has been designed to give a range of movement and adjustment positions while sitting on the lifting blades or tines in or out, this is to maximise the length on blades to accommodate two planks. Furthermore, the design of the hole pattern is so that once a user has used the centre point device **100**, while a bike is mounted on the plank **32**, the user is able to mark the centre point on the plank **32** then using the measuring device component which equals 6×87 mm, the distance between each hole on the plank longitudinal is 522 mm which is the same distance as that between the holes **25** drilled and tapped in the lifting blades **24**. This means there are many securing points while lifting the bike to ensure a balanced weight distribution while elevated. The different degrees outlined in the drawings are done so that the holes will fit on the plank leaving spacing on the outer edges to accommodate a bolt so as not to hit the stiffener walls of the plank **32**.

(76) As illustrated in FIG. **29**, the 12 mm holes **92** are stamped and angled upwardly or inwardly that is slightly raised by around 3 mm. This is done to trap any small amounts of fluids, oils or fuel on the horizontal portion of the plank **32** and to inhibit them from draining onto the bike/vehicle underneath, to thereby protect them from damage. The reader will appreciate that the hole illustrated in FIG. **29** is positioned on the vertical end stop **80**, however they will appreciate that the holes **32** on the horizontal portion of the plank **32** are similarly configured.

(77) FIG. **30** illustrates an alternate embodiment of the ground engaging members **16**, having rollers **18** comprise a pair of wheels **132**, **134** mounted to an axle **136**. The wheels **132** are retained within respective mounts **138** at the ends of the members **16**. The wheels **134** are positions on outer sides of members **16** and are provided as outriggers to improve stability of the lifting jack **12**.

(78) The dolly **30** can also be used on its own to store and move the vehicle being stored thereon, wherein the vehicle, such as a bike can be rotated horizontally around 360 degrees with the bike securely attached to the dolly **30**.

(79) FIGS. **31** to **34** illustrate another embodiment of the spaced apart lifting tines **24**, illustrating the position of the nuts **140** for securing the plank **32** thereto. FIG. **34**, further illustrates the positioning of two planks **32***a* and **32***b* on a single pair of lifting tines **24**, for holding respective bikes thereon in a side by side arrangement. The skilled addressee will now appreciate the advantages of the illustrated invention over the prior art. In one form the present invention provides a means for storing a vehicle, such as a motorbike, at a height above the ground. As the reader will appreciate, this has significant advantage where space is at a premium, such as in a car garage or storage locker, or where multiple bikes are being stored. The reader will appreciate that the apparatus can be used to store other vehicles, such as but not limited to, jet skis and other personal watercraft (PWC).

(80) Various features of the invention have been particularly shown and described in connection with the exemplified embodiments of the invention, however it must be understood that these particular arrangements merely illustrate the invention and it is not limited thereto. Accordingly, the invention can include various modifications, which fall within the spirit and scope of the invention.

## Claims

1. A vehicle storage apparatus, including: a lifting jack having spaced apart lifting tines; a dolly being reversibly couplable to the lifting tines in a number of alternate positions; and a centre point device for determining a weight distribution along a longitudinal axis of the dolly when a vehicle is located thereon, to assist in a selection of one of said number of alternate positions for a suitable attachment of the dolly to the lifting tines, to thereby inhibit tipping of the lifting jack when the lifting tines are in a raised position or otherwise dislodgement of the vehicle stored thereon, wherein the dolly being movable laterally relative to the lifting jack and reversably couplable to the lifting tines, once a longitudinal centre point of the dolly and vehicle is determined.
2. The vehicle storage apparatus in accordance with claim 1, wherein the dolly being couplable to the lifting tines, using a nut and bolt that are engageable through co-axially alignable primary holes, or a clamp mechanism or a clip may be provided such that the dolly can be coupled to, or couplable to the lifting jack in a variety of positions depending upon the weight distribution of the vehicle being supported on the dolly.
3. The vehicle storage apparatus in accordance with claim 2, wherein a guide member is slidably coupled to or configured to engage with the dolly and being used in conjunction with the centre point device to determine which of a plurality of primary holes extending through the dolly are to be used to coaxially align with a secondary hole through one of said lifting tines.
4. The vehicle storage apparatus in accordance with claim 3, wherein the dolly plan includes an upwardly open channel and the guide member clips over an edge of the upwardly open channel and being configured to slide therealong.
5. The vehicle storage apparatus in accordance with claim 2, wherein the centre point device comprises a roller having ends being enlarged to abut or be positioned adjacent respective sides of the dolly, to thereby inhibit the centre point device from rolling sidewardly out from under the dolly when in use.
6. The vehicle storage apparatus in accordance with claim 1, wherein the dolly and centre point device are separate components.
7. The vehicle storage apparatus in accordance with claim 1, wherein the dolly and centre point device are unitary in construction.
8. The vehicle storage apparatus in accordance with claim 1, wherein the lifting jack may be a mobile unit having lockable caster wheels, ground engaging members with rollers, a vertical frame and lifting mechanism for raising the spaced apart lifting tines, or the lifting jack comprises a frame attachable to a wall or other vertical support, and lifting tines movably connected thereto.
9. The vehicle storage apparatus in accordance with claim 1, wherein the dolly comprises a plank being affixed to two spaced apart cross members, each cross-member includes a caster wheel assembly or slider at each end thereof, such that the plank is effectively supported on four caster wheels or sliders, each cross-member further including tie-down eyelets for an attachment of ratchet straps or the like, to assist in securing the vehicle when being stored.
10. The vehicle storage apparatus in accordance with claim 9, wherein the plank includes a detachable ramp at a first end, and a wheel engaging member at an opposite second end, adjacent a vertical end stop.
11. A method of storing a motorbike or vehicle, including a plurality of steps of: providing a vehicle storage apparatus in accordance with claim 1; positioning the motorbike or vehicle on the dolly; lifting the dolly using the lifting tines of the lifting jack; positioning the centre point device under the dolly; lowering the dolly using the lifting tines of the lifting jack until the dolly rests on the centre point device; moving the dolly in a longitudinal direction to determine generally the centre point or fulcrum of the motorbike or vehicle and dolly, such that the weight is even distributed on either side of the centre point device; marking or otherwise noting the centre point or

fulcrum and other related measurements on the dolly; lifting the dolly off the centre point device using the lifting jack; removing or moving the centre point device; and using the markings of the centre point or other related measurement, to fix the dolly to the lifting tines of the lifting jack, such that when the dolly and motorbike or vehicle are raised by the lifting jack the vehicle storage apparatus is inhibited from tipping or otherwise dislodgement of the motorbike or vehicle stored thereon.

12. The method in accordance with claim 11, further including a step of securing the motorbike or vehicle to the dolly, using at least one wheel engaging member, and/or clamps, and/or straps and/or other reversible attachable member for holding the motorbike or vehicle in place relative to the dolly.

13. The vehicle storage apparatus in accordance with claim 1 being a motorbike storage apparatus.

14. The vehicle storage apparatus in accordance with claim 13, including: the dolly having a plurality of generally vertical primary holes extending therethrough, the dolly being supportable on wheels or slides, and configured to hold a motorbike thereon; the lifting jack including spaced apart lifting tines, each of the lifting tines including at least one generally vertical secondary hole extending therethrough; and the centre point device, being configured to selectively support the dolly thereon in a number of alternate positions, for use in determining the weight distribution along a longitudinal axis of the dolly, whereby at least two of the vertical primary holes are identified for coaxial alignment with respective secondary holes extending through the lifting tines, to thereby enable passage of a respective fixing means therethrough, wherein the dolly is fixedly connectable to the lifting tines to thereby provide a generally even distribution of weight to inhibiting tipping of said apparatus or otherwise dislodgement of the motorbike stored thereon.